

# DRRR

## Disaster Readiness and Risk Reduction

Grade 12, Semester 1 study outline

*A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.*

---

### 01. Introduction to Disasters

#### Introduction to Disasters

##### INTRODUCTION

- Disasters have always been the outcome of interactions between humans and other living things, technology, and the natural world.
- The way we live our daily lives is constantly impacted by many forms of disasters, which might be immediate, unanticipated, or slow-moving.

##### DISASTERS

- A DISASTER is an event which causes loss of lives, negative impact to health status, and great damages to properties and the environment.
- It is a major disruption of a community's functioning and almost always includes widespread human, material, or environmental losses.

##### TYPES OF DISASTERS BASED ON CAUSE

###### 1) NATURAL DISASTERS

- Examples of Natural Disasters include Earthquakes, Volcanic Eruptions, Floods, Storms and Plagues.
- These are almost always viewed in relation to their magnitude or intensity and speed of onset.

###### 2) MAN-MADE DISASTERS

- Examples of Man-made disasters are War, Pollution, and Industrial Accidents (Oil and Chemical Spills, etc.)

##### TYPES OF DISASTERS BASED ON ONSET

###### 1) RAPID ONSET

- Examples of Rapid Onset Disasters are Flooding (caused by Typhoons) and Earthquakes. -- 1 of 2 --

###### 2) SLOW ONSET

- Examples of Slow Onset Disasters are thawing of permafrost (caused by the melting of the ice caps) and Drought.

##### THE NEED TO STUDY DISASTERS

- There are primarily two reasons why Filipinos should study disasters: 1) We live in a disaster-prone area.
- 2) We suffer the consequences of these disasters.

##### DISASTER RISK

- There are two types of disaster risk: 1.
- Intensive risk is defined as the risk connected to the exposure of significant populations and economic activity to intense hazard events that have the potential to have devastating effects on human life and property.
- Extensive risk is the widely dispersed risk associated with exposing dispersed populations to recurring or persistent low- to moderate-intensity hazard circumstances, frequently of a very localized type, which can have catastrophic cumulative disaster effects.

##### IMPACTS OF THE DISASTER

- Direct Losses - include physical losses such as lives and damages (to infrastructures) or impairments (to persons).

##### 2. Indirect Loses - Includes loss of income and increased cost of basic necessities

- Intangible Losses- Include psychological impact such as depression and other similar conditions Shariff A .

##### Key terms

Examples of Man-made disasters - War, Pollution, and Industrial Accidents (Oil and Chemical Spills, etc.)

Disasters - also classified based on how sudden they may occur and how immediate their impact are

## 02. Introduction to Exposure and Vulnerability

### Introduction to Exposure and Vulnerability

#### INTRODUCTION

- Last lesson, we discussed what disasters are and why Filipinos should study Disaster Readiness and Risk Reduction.
- To further understand Disaster Risk, it should be understood in conjunction with the concepts of hazard, exposure, vulnerability, and sensitivity.

#### HAZARD

- Hazard is a process, phenomena, or human activity that could result in a loss of life, an injury, other health effects, property damage, social and economic upheaval, or environmental degradation is referred to as a hazard.

#### EXPOSURE

- It is defined as the number and status of people, infrastructure, housing, production capacity, and other tangible human assets located in hazard-prone places.
- The quantity of people or the sorts of assets in a region are just two examples of indicators of exposure.

#### VULNERABILITY

- Vulnerability is "the characteristics of a person or group and their situation that influences their capacity to anticipate, cope with, resist, and recover from the impact of a natural hazard." The types of vulnerability are as follows: 1) Social Vulnerability - This vulnerability occurs because of societal rules, t...

#### 2) Psychological Vulnerability - This vulnerability occurs because of trauma of loss

#### 3) Physical Vulnerability - This vulnerability occurs because of lack of assets

- To further understand the interplay between exposure and vulnerability, another concept must be understood.
- This concept is called Sensitivity.
- The possible damage or harm to people and assets brought upon by the interplay of exposure and vulnerability is called SENSITIVITY.
- As a result, these various consequences and outcomes of catastrophes must also be taken into consideration by policies intended to manage risk and vulnerability.

#### VULNERABILITY ASSESSMENT

It is a method of determining a specific community's

#### 1) Susceptibility to hazards (Exposure); and

#### 2) Resilience to cope (Vulnerability) and withstand the impacts of hazards (Sensitivity)

- It must include the following:

#### 1) Hazards and their

- -- 1 of 2 -- a) Potential Characteristics b) Frequency c) severity

#### 2) Geographical areas (and communities therein) that are most susceptible to hazards

#### 3) Sectors of the community that would be affected by each hazard

#### 4) Assessment of a community's ability to withstand and cope with the effects of a hazard

- An example of the contents of a Vulnerability Assessment in a more contextualized setting will be provided in the next lesson.

#### 1) It is the center of a disaster preparedness plan

#### 2) It shows the geographic location where disasters may have the worst impact

#### 3) It shows which community requires priority in terms of assistance and aid

#### 4) It provides input to the national government in terms of disaster preparedness strategy

#### Key terms

Hazard - process, phenomena, or human activity that could result in a loss of life, an injury, other health effects  
Different populations - exposed to varying degrees of risk caused by Hazards. Vulnerability is "the characteristics  
Wakandans - EXPOSED but not VULNERABLE, thus they are not SENSITIVE to the HAZARD.

---

## 03. Contextualizing Exposure and Vulnerability

### Contextualizing Exposure and Vulnerability

#### INTRODUCTION

#### HAZARDS IN THE PHILIPPINES

- Typhoons and large waves are the causes of storm surge.
- Damage to property includes damages to agriculture, infrastructures, and properties on both public and private land.

#### THE ECONOMIC COST OF DISASTERS IN THE PHILIPPINES

#### THE HUMAN COST OF DISASTERS IN THE PHILIPPINES

- 2011 came next, the year that saw Tropical Storm Sendong impact the nation in December.
- 12,097 deaths were reported in the Philippines from 2010 to 2019. -- 1 of 3 --

#### PREPAREDNESS PLANNING

- These proactive steps are called Preparedness Plans. Its goal is the reduction of loss of life and livelihood.

#### COORDINATION

- After Preparedness Planning, coordination in the implementation of plans must be ensured.
- There are primarily two types of coordination:

#### 1) Vertical - One of the parties to the coordination process has more authority

- 2) Horizontal - The parties involved in the coordination process have the same authority or are operating at the same level.

#### WHAT IS A VULNERABILITY ASSESSMENT?

- A Vulnerability Assessment must include:

#### 1) Hazards and their

#### 2) Geographical areas (and communities therein) that are most susceptible to hazards

#### 3) Sectors of the community that would be affected by each hazard

#### 4) Assessment of a community's ability to withstand and cope with the effects of a hazard

#### Low-lying Areas, Catch Basin)

---

## 04. Hazards

### Hazards

#### INTRODUCTION

- A Hazard is a dangerous phenomenon, object, or behavior that may result in wrongful death, bodily harm, other health effects, property damage, loss of jobs and services, disruption of the social and economic system, or damage to the environment.

#### NATURAL HAZARDS

#### 1) Biological

#### 2) Geologic

#### 3) Hydrometeorological

#### Biologic Hazards

- Some examples of biological hazards are: 1) Serious illness outbreaks including avian influenza and severe acute SARS, often known as the Middle

#### East respiratory illness (MERS)

#### 2) Pestilences and infestations caused by insects or other animals, etc

- 3) Exposure to biological hazards due to unintentional discharge of microbes from places like research facilities, or by conscious actions 4) Viral, bacterial, and other major epidemic and possible pandemic diseases (yellow fever, cholera, zika virus, Ebola, bacterial, fungal, and prion illnesses) and those based on...

#### Geologic Hazards

- Some examples of Geologic Hazards are: 1) Earthquakes: Earthquakes refer to sudden slip on a fault as well as the resulting ground shaking and seismic energy emitted from the slip, as well as to sudden stress changes in the earth induced by volcanic or magmatic activity.

## 2) Landslide/debris flows: Movement of soil, rocks, and other debris down a slope

- 3) Tsunami: A tsunami is a sea wave that can have a local or remote source and is caused by large-scale seafloor displacements brought on by powerful earthquakes, significant underwater slides, or erupting volcanic islands.
- 4) Volcanic eruption: This refers to the explosive release of gasses, lava, and fragmented ejecta from a volcanic vent.
- 5) All other volcanic activities like rock fall, ash fall, lava streams, gasses etc.

## Hydrometeorological Hazards

- Examples of Hydrometeorological Hazards are: 1) Tropical Cyclones: a system of confined low pressure in the atmosphere that rotates in the Northern Hemisphere counterclockwise and clockwise in the Southern Hemisphere (includes cyclones, extra- -- 1 of 2 -- tropical cyclones, tropical cyclones, hurricanes, and typhoo...
- 2) Floods: the exceeding of a stream's or other body's typical boundaries of water, or the buildup of water where it would not ordinarily be submerged.
- 3) Storm Surge: transient elevational rise of the sea that occurs temporarily in a certain location because of extremely bad weather (high winds and/or low air pressure).

## HUMAN-INDUCED HAZARDS

### 1) Technology-induced

### 2) Terrorism

#### Technology-induced Hazards

- Examples of Technology-induced hazards are: 1) Man-made Hazards: Hazards that are "induced entirely or predominantly by human activities and choices 2) Nuclear Emergencies: These are caused by nuclear chain reactions.
- 3) Radiological Emergencies: These can start from radiation sources, including radiography equipment, radioactive material used in industry, missing sources, and more.

#### Terrorism

- This refers to the purposeful use of violence to instill a culture of terror among a community and achieve a specific political goal.

## SOURCES

---

# 05. Earthquake Hazards

## Earthquake Hazards

### INTRODUCTION

- Being located in the "Pacific Ring of Fire," the Philippines is vulnerable to various types of natural hazards due to its geographic location and physical environment.
- Thus, it is imperative for Filipinos to have an adequate knowledge of the nature and cause of Earthquakes.

### EARTHQUAKES

- The theory of Plate Tectonics, which combines the concepts of continental drift and sea-floor spreading, proposes that the earth's surface is made up of several relatively thin (100-150 km), rigid segments or plates that move continuously in relation to one another over a semi-plastic layer (asthenosphere) underneath...
- The fault or fault plane is the area where they slide.
- The epicenter is the point on the earth's surface that is directly above the hypocenter, which is where the earthquake begins under the surface.
- Before the greater earthquake occurs, scientists are unable to determine whether an earthquake is a foreshock.
- The mainshock is the term used to describe the greatest earthquake.
- Aftershocks that follow mainshocks are a given.

### TYPES OF FAULTS

- 1) DIP-SLIP FAULTS are faults where two blocks move along the direction of the dip plane.

### 3) OBLIQUE-SLIP FAULTS are faults where two blocks move both horizontally and vertically

### TYPES OF EARTHQUAKES

- There are two types of this kind of earthquake and it depends on what caused it.

### 1) TECTONIC EARTHQUAKES are caused by abrupt movement of the tectonic plates along the faults

### 2) VOLCANIC EARTHQUAKES are caused by volcanic eruptions

- To further understand the impact of earthquakes, we should first understand how earthquakes are measured.

## MEASURING EARTHQUAKES

### 1) Magnitude - The energy released at the epicenter of an earthquake is measured in magnitude

- Measurements made with seismographs are used to calculate magnitude.
  - The distance to the epicenter must also be taken into consideration because the waves become smaller the more away you are from the earthquake.
  - 2) Intensity - The strength of the shaking caused by an earthquake at a certain site is measured by intensity.
  - The Modified Mercalli earthquake intensity scale is the one that is most frequently used.
- 

## 06. Earthquake Readiness

### Earthquake Readiness

#### INTRODUCTION

- The Philippine fault zone (PFZ), which spans 1,200 km, is a significant tectonic feature that spans all of the Philippine archipelago, running from Mindanao in the southeast to northwest Luzon.
- In order to mitigate the impact of an earthquake, there are various concepts that must be understood.
- One of these concepts is the concept of SIGNS.

#### SIGNS OF AN IMPENDING EARTHQUAKE

- In order to lessen the impact of earthquakes, communities that are exposed and vulnerable to earthquakes must look for signs of an impending earthquake.
- The following are the most common signs: 1) Animal behavior - A severe earthquake was predicted to occur a few days after rats, weasels, snakes, and centipedes reportedly fled their homes in search of safety.
- There is a wealth of anecdotal data that suggests odd behavior in animals, fish, birds, reptiles, and insects weeks to seconds before an earthquake.
- 2) Atmospheric changes - While there is no consensus in the scientific community, abrupt changes in the atmosphere and the existence of atmospheric lights MAY be used to predict earthquakes.
- 3) Electromagnetic disturbance - Registration of disturbances in the atmospheric electromagnetic field may be used to predict earthquakes in the short term.

### 4) Boom sound- Before an earthquake, a loud bang similar to artillery fire may be heard

#### EARTHQUAKE HAZARDS

- In order to further prepare for Earthquakes, communities must also identify the hazards associated with earthquakes.
- Ground shaking and its impact are the first main earthquake hazard.
- Tsunamis are a succession of enormous waves brought on by an undersea volcano or earthquake that abruptly moves the seafloor.
- A Sinkhole is a rare earthquake hazard that can happen at any moment.

#### EARTHQUAKE READINESS

##### 1) Arrange a disaster or emergency supply kit which should include

##### 2) Make an earthquake preparedness plan which should include

##### 3) Discuss the earthquake preparedness plan with your family members

##### 4) Secure and be aware of possible hazards related to earthquakes

#### Key terms

Soil Liquefaction - also an earthquake hazard. When liquefaction takes place, buildings may potentially sink into  
Surface Rupture - another dangerous earthquake hazard. It occurs when movement on a fault deep within the  
Tsunamis - succession of enormous waves brought on by an undersea volcano or earthquake that abruptly  
Fires - also hazards that may be triggered by earthquakes. The collapse of infrastructures may also cause the

---

## 07. Volcanoes

### Volcanoes

#### INTRODUCTION

- A Volcano is a hole or rupture in the surface or crust of a planet which permits hot magma, volcanic ash, and gasses to escape from the magma chamber below the surface.
- Instances of volcanoes resulting from divergent tectonic plates breaking apart can be found along a mid-oceanic ridge, such as the Mid-Atlantic Ridge, whereas examples of convergent tectonic plate movement can be found along

the Pacific Ring of Fire.

- Alternatively, there is a solid mass of rock termed the mantle deep within Earth, sandwiched between the molten iron core and the thin crust above the surface.

## CONCEPTS

- Higher viscosity means a fluid will flow very slowly.
- 2) Lava - molten or partially molten rock that has exited from underground to the surface of the earth.

### 3) Magma - hot liquid and partially molten rock located under the surface of the Earth

- 4) Volcanic Gas - gasses that exist under the surface of the Earth.
- 5) Tephra - these are pyroclastic materials (mixtures of molten rocks, gasses, dust, etc) made airborne by a volcanic eruption.

## TYPES OF VOLCANOES

- The type of volcanic material that each volcano produces varies, and this has an impact on the volcano's size, form, and structure.
- 1) Shield Volcano - This type of volcano forms when a surface vent spews low viscosity, runny lava.
- Examples of shield volcanoes are Mauna Kea and Mauna Loa, both found in Hawaii.
- 2) Stratovolcano - These volcanoes are formed when difficult-to-flow lava that is viscous and sticky flows from the crater.
- 3) Calderas - These volcanoes are formed when the topmost portion of a volcano is blown off by an explosive eruption.

### 1) Active Volcanoes - these are volcanoes which erupted in the last 10,000 years

- 2) Dormant Volcanoes - these are volcanoes that have never erupted in the last 10,000 years, but are expected to erupt again in the distant future.
- 3) Extinct Volcanoes - there are volcanoes that have never erupted in the last 10,000 years, but are never expected to erupt in the future. -- 1 of 2 -- Take note that even though no one expects an extinct volcano to erupt, there is still a chance that an extinct volcano may become active again.
- Example of a previously thought extinct volcano is the Pinatubo volcano.

---

## 08. Volcanic Disaster Readiness

### Volcanic Disaster Readiness

#### INTRODUCTION

- We have also discussed the different types of volcanoes.

#### SIGNS OF AN IMPENDING VOLCANIC ERUPTION

- 1) Audible earthquakes
- 2) Changes in the temperature
- 3) Changes in the chemical composition of groundwater and spring fluids
- 4) Alterations on the ground surface
- 5) Steam-blast eruptions, which happen when superheated water flashes to steam

#### TOOLS FOR RECOGNIZING VOLCANIC HAZARDS

##### 1) Permanent Danger Zones (PDZ) - PHIVOLCS (Philippine Institute of Volcanology and Seismology)

- 2) Alert Levels - similar to the Permanent Danger Zones, Alert Levels vary per volcano.
- Furthermore, specific parameters must be achieved to remove an alert level.

#### VOLCANIC HAZARDS

- It is a major threat to vegetation and infrastructures, but not to humans.
- 2) Pyroclastic Flow and Surges - This occurs when rock materials, hot gasses, and ash flow and may result in burning and suffocating people.
- 3) Gas Emissions - This occurs when minor amounts of nitrogen, methane, and helium from magma and lava are blown by wind, causing damage to the health of affected communities.
- 4) Ash Fall - These are small pieces of rocks, minerals, and fragments blown by a volcanic eruption.

##### 5) Lahar - These are mudflows which are a combination of volcanic debris and floods

#### SURVIVING VOLCANIC ERUPTIONS

##### 1) Make a detailed emergency plan. This must

- a) Identify Safe Zones b) Identify Evacuation Centers c) Identify Evacuation Routes d) Ensure means of transportation e) Stay away from rivers and streams.

## 2) Arrange an Emergency Supply Kit, which must include the following items

- Examples of such documents are: a) Birth Certificate b) Medical Records c) Passports d) Identification Cards 4) Develop communication mechanisms.

## 5) Monitor News Updates and Coordinate with Local Authorities

### Key terms

While our municipality - not directly exposed to a volcanic hazard, the following information will give us an

---

## 09. Other Geologic Hazards

### Other Geologic Hazards

#### INTRODUCTION

##### 1) Sudden Geologic Hazard

##### 2) Slow Geologic Hazard

#### SUDDEN GEOLOGIC HAZARD (SGH)

##### 1) Small-scale (low exposure) - this means that SGHs scope of potential damage is very limited

- 2) Low-severity (low vulnerability/sensibility) - the possible impact of SGHs are relatively small.
- 3) Localized (community-based) - Since SGHs's scope is limited, it can only affect a small number of communities.

##### 4) They usually take the form of landslides

- 2) Any other action that may cause unstable, elevated soil to move downhill (explosions, excavations, earthquakes, etc.).

#### LANDSLIDES

- Landslide refers to all types of surface, downhill, movements of soil, rock, or debris, caused by gravity.

#### LANDSLIDES BASED ON DEPTH

- There are two types of landslides based on depth: Shallow Landslides and Deep-Seated Landslides.
- The main difference between this kind of landslides is the depth of soil, rocks, or debris that was moved.
- If the depth is less than 10 meters, it is called a Shallow Landslide.
- If it is more than 10 meters, it is called a Deep-seated Landslide.

#### LANDSLIDES BASED ON MOVEMENT OF MATERIALS

- There are four types of landslides based on movement: 1) Fall - This occurs when soil or rocks abruptly detach from a highly elevated, steep slope or cliff (about 45 to 90 degrees).
- 2) Topple - This occurs when the bottom part of a geologic feature is not solid enough to support the mass above, resulting in the upper part toppling down.
- This kind of movement usually occurs around a point or near the base of a landform.
- 3) Slide - This occurs when the upper portion of a geologic feature is separated from its downslope bottom part.
- This kind of landslide is usually deep-seated.
- 4) Spread - This occurs when the top portion of a geologic feature moves or spreads into a weaker lower portion, usually because of liquefaction.
- 5) Flow - This is a phenomenon where displaced rocks (rock flow), debris (debris flow), or soil (earth flow) "flows" continuously due to slopes or presence of water.

### Key terms

Sudden Geologic Hazards - hazards occurring on the Earth's surface that happen very quickly with little or no  
Landslide - all types of surface, downhill, movements of soil, rock, or debris, caused by gravity. It may be

---

## 10. Other Geologic Disaster Readiness

### Other Geologic Disaster Readiness

#### INTRODUCTION

#### SLOW GEOLOGIC HAZARDS

- They are two examples of Gradual Hazards:

### 1) Subsidence

### 2) Alluvial Fan

#### SUBSIDENCE

- Land Subsidence is the gradual sinking of surface lands due to any of the following factors:

### 1) Slightly soluble rocks

### 2) Percolating Water

### 3) Water Table

### 4) Cave Roof

#### Suffosion

- It is generally known that these suffocating characteristics are hazardous to infrastructure and buildings, both during construction and afterward.

#### ALLUVIAL FANS

#### OTHER GEOLOGIC HAZARD DISASTER READINESS

- Thus, to avoid loss of life and property, signs must be taken seriously as they may herald incoming geologic disasters.

### 1) Infrastructure Signs

### 2) Ground Signs

### 3) Ground Signs

#### Key terms

Land Subsidence - gradual sinking of surface lands due to any of the following factors:

---

## 11. Hydrometeorological Hazards

### Hydrometeorological Hazards

#### WHAT IS HYDROMETEOROLOGY?

- Hydrometeorology is the study of the interactions between the atmospheric, hydrospheric, and terrestrial phases of the hydrological cycle (i.e. the transfers of water and energy between the land surfaces, bodies of water, and the lower atmosphere).

- It is composed primarily of three distinct but related branches: 1) Atmospheric Science - This branch aims to understand and predict the mechanisms that lead to our weather by focusing on the interplay between variables in the Earth's atmosphere.

- Examples of such variables are: a) Temperature b) Pressure, c) Water/Water Vapor 2) Hydrology - This branch studies the movement, distribution, and quality of water on earth.

### 1) Climatology - the science that studies the historical weather patterns of our planet

- 2) Cloud Physics - the science that deals with the processes involved in the formation and growth of clouds.

#### WHAT ARE HYDROMETEOROLOGICAL WEATHER SYSTEMS (HMS)?

- The following are hydrometeorological weather systems that affects or impacts the Philippines: 1) TROPICAL CYCLONE (BAGYO) - It is a rapidly rotating system characterized by a low-pressure center, a closed atmospheric circulation, strong winds, and thunderstorms that produce heavy rain and squalls.

### 2) SOUTHWEST MONSOON (HABAGAT) - It is composed of moisture-laden winds and rains and comes in

- the form of heavy rains, strong winds, and flooding.

### 3) NORTHWEST MONSOON (AMIHAN) - It is composed of winds from northern Asia and has moderate

- 4) INTERTROPICAL CONVERGENCE ZONE (ITCZ) - It is a low pressure area with high humidity which forms a band of clouds across the equator.

- It is the result of northeasterly and southeasterly winds.

#### COMPARISON OF HMS

#### SOURCE NATURE HAZARDS

#### BAGYO

Pacific Ocean,

West Philippine



Sea  
Event  
Thunderstorms,  
Strong winds,  
Heavy rains  
HABAGAT  
Hot, moist wind  
Asia  
Season Strong winds,  
Heavy rains  
AMIHAN Icy winds from  
North Asia Season Little to no rain  
ITCZ  
Convergence of  
Northeasterly and  
Southeasterly  
Event/  
Season Heavy rains  
Notable Examples of HMS-related incidents in the Philippines  
Key terms

Hydrometeorology - study of the interactions between the atmospheric, hydrospheric, and terrestrial phases

---

## 12. Hydrometeorological Disaster Readiness

### Hydrometeorological Disaster Readiness

#### INTRODUCTION

- As such, we must be prepared in order to mitigate possible impacts of these HMS.

#### HYDROMETEOROLOGICAL HAZARDS

- The following are hazards related to Hydrometeorological Weather Systems: 1) Lightning - It is a bolt of electricity, often blue-white in color, which travels from cloud to cloud, or from cloud to the surface of the earth.
- 2) Hail - It is a form of precipitation which comes as balls or irregular lumps of ice.
- It is a common occurrence on top of mountain ranges in the Philippines.
- 3) Tornado - It is a violently rotating funnel-shaped cloud which descends from a thundercloud to the ground.
- 4) Strong Winds - These are winds brought upon by tropical cyclones, and may be an onset of tornadoes.
- 5) Floods and Flash Floods- This refers to large amounts of water covering a typically dry land area, usually brought upon by heavy rains and tropical cyclones.
- If a flood is caused by sudden release of dams or flows in a river (which causes fast currents), it is called a flash flood.
- 6) Storm Surge - This occurs when large waves are created because of rising waters caused by low pressure areas.

#### HYDROMETEOROLOGICAL TOOLS

- Examples of such tools are:

##### 1) NDRRMC Weather Situation Site - this site can help monitor possible hazards brought by HMS

- 2) NDRRMC Flood Advisories Site - this can help monitor floods by showing possible areas where flooding can occur based on a current HMS phenomenon.
- 3) Typhoon Signals - Typhoon signals provide a clear picture of what a community can expect (including Typhoon Strength) as regards an incoming typhoon.
- 4) Rainfall Advisory Systems - This provides a color-coded advisory as regards the amount of rain a community may experience.

#### HYDROMETEOROLOGICAL DISASTER MITIGATION

##### GENERAL TIPS

##### 1) Stay updated on hydrometeorological hazards

##### 2) Stay alert and be on the lookout for possible floodings, tornadoes, and surges

## **DURING THUNDERSTORMS**

- -- 1 of 2 -- 1) Seek safe shelter.
- 2) Stay away from corded phones, windows, and doors or those connected to plumbing, electrical wires.

### **3) Unplug all electrical equipment before the thunderstorm occurs**

## **DURING TORNADOES OR STRONG WINDS**

- 1) Stay indoors. Flying debris can cause injuries or even death
- 2) Stay out of auditoriums, gymnasiums, and other similar structures
- 3) If caught outside and have no time to escape, lie flat in the nearest ditch or ravine
- 4) When inside a car, go outside and find a sturdy structure as shelter

## **BEFORE FLOODS AND STORM SURGES**

- 1) Find out and be informed of your community's evacuation plan
- 2) Know possible emergency enters
- 3) Prepare a list of emergency contact numbers in case of severe storms and heavy floodings
- 4) Prepare first aid kit and emergency provisions
- 5) Clear debris blocking or clogging your gutters and downspouts

### **Key terms**

Since the Philippines - prone to hydrometeorological hazards, the government developed tools that Filipinos may

---

## **13. Fire Hazards**

### **Fire Hazards**

#### **INTRODUCTION**

- This, together with earthquakes and typhoons, is a fairly common occurrence in the Philippines.
- As such we must learn to prepare and mitigate its impact in our communities.

#### **FIRE DISASTERS**

- 2) Natural Fires - These are caused by natural processes, such as lava flows, lightning strikes, and earthquakes.

#### **WILDFIRES**

- Some scientists believe that wildfires are a necessary process in order to bring life to forests.

- 1) They destroy the homes of people who reside within or near forests
- 2) It can decimate forest ecosystems
- 3) It can destroy trees
- 4) It destroys the natural habitats of animals, especially those that are endemic in the area

#### **FIRES AND CLIMATE CHANGE**

- 1) It accelerates the drying of organic materials (the material that burns and spreads fire)
- 2) It decreases moisture content of flammable materials causing it to catch fire easier
- 3) It creates hotter days and longer summer seasons

#### **ADDRESSING FIRES**

- The most commonly used is the FIRE ALARM SYSTEM.
- They differ on the 1) estimated number of infrastructures affected, the estimated number of fire trucks required, and the fire ground commander that must respond.
- The following are the Fire Alarm Levels:

#### **FIRE ALARM**

##### **LEVEL**

##### **ESTIMATED**

##### **NUMBER OF**

##### **AFFECTED**

##### **INFRASTRUCTURE**

##### **ESTIMATED**

##### **FIRE TRUCKS**

##### **NEEDED**

## **FIRE GROUND COMMANDER**

### **First Alarm 2-3 Houses 4 Fire**

#### **Trucks**

#### **Team Leader**

#### **Second**

### **Alarm 4-5 Houses 8 Fire**

#### **Trucks Senior Fire Officer**

#### **Third**

#### **Key terms**

Fire Disasters - events involving fire which caused death and/or widespread damages to a community or section

Wildfires - caused by the ignition of combustible vegetation (dry leaves, woods, etc.) due to hot dry climates

---

## **14. Fire Disaster Readiness**

### **Fire Disaster Readiness**

#### **INTRODUCTION**

#### **THE FIRE TRIANGLE**

- The three elements required to start and maintain a fire are known as the fire triangle.
- It is composed of Heat, fuel, and oxygen.
- 1) HEAT - A heat source is necessary for ignition to take place, and various materials have various "flash points" or points when they will ignite.
- Water application can reduce the heat produced by various types of fire.
- 2) FUEL - These are materials that have low flash points.
- Fuels should be stored properly to avoid becoming a fire hazard because they are likely the hardest "side" of the fire triangle to extinguish.
- 3) OXYGEN - Since oxygen combines with burning fuel to release heat and CO<sub>2</sub>, it is necessary to maintain the combustion reaction.

#### **FIRE HAZARDS**

- The following are fire hazards: 1) APPLIANCE/EQUIPMENT - Since these are used daily by people, there are also higher chances that they generate heat.

#### **2) ARSON - These are deliberate and criminal actions where a person maliciously ignite (heat) a**

- 3) CANDLES/FIREWORKS/CIGARETTES - the careless use of these materials (heat) may cause fire disasters if their heat reaches a source of fuel.

#### **HOW TO AVOID FIRES**

- In order to avoid fires, we should remember and do the following measures: 1) Observe pre-disaster mitigation and preparations to avoid fire.

#### **2) Follow a fire safety plan at home. This shows the location of**

- These fire hydrants are used by firefighters to stop fire disasters.
- 4) Make sure that the building you are in or you regularly visit has fire alarms and fire sprinklers.

#### **WHAT TO DO WHEN THERE IS A FIRE?**

- 1) Always use the stairs, not elevators.
- 2) Never re-enter a burning building, unless allowed by an authority from the fire department.

#### **4) Assist the disabled or people with special needs in exiting the building**

- 5) Shut down doors to isolate the area where the fire started, thus slowing its spread.
- 

## **15. Disaster Risk Reduction**

### **Disaster Risk Reduction**

#### **INTRODUCTION**

- Disaster Management includes all types of operations, including structural and non-structural precautions, necessary to prevent or reduce the negative impacts of disasters and their hazards.

- As such, it has always emphasized the preparedness for response during a catastrophe or a disaster.

#### **WHAT IS DISASTER RISK REDUCTION (DRR)?**

- Disaster Risk Reduction is the concept and practice of reducing disaster risks.
- It is done through: 1) Analysis of Hazards - this is the process where a hazard is broken down into analyzable segments in order to a) determine its impact to a community, b) Identify possible scope or range of damage, c) how it might cause damage.
- 2) Management of Causal Factors of Disasters - this is the manner by which Disaster Risk Reduction actors try to minimize and control possible causes of some disasters.
- This is an example of management of causal factors of disasters.

#### **WHAT DOES DISASTER RISK REDUCTION LEAD TO?**

- Reduced exposure to hazards - Remember that EXPOSURE is the number of people or assets that may be affected by hazards.

### **3. Effective management of land and the environment with proper management - knowing when a**

#### **WHAT ARE KEY PRINCIPLES OF DRRM?**

- -- 1 of 2 -- Key Principles of DRRM provide the following: 1.
- It is an AID that makes communities understand why disaster plans must be prioritized. (e.g., minimize refusal to adhere to safety protocols) Now that we know the benefits of the key principles of DRRM, the following are the key principles of DRRM.

#### **WHAT ARE KEY PREMISES OF DRRM?**

##### **Key terms**

Disaster Risk Reduction - concept and practice of reducing disaster risks. It is done through:

## **16. Community Programs for Disaster Risk Reduction and Readiness**

### **Community Programs for Disaster Risk Reduction and Readiness**

#### **INTRODUCTION**

- Community is a group of people living together in the same area and as such exposed and vulnerable to the same set of hazards.

#### **CPDRRR**

- It is a process by which at-risk communities are actively engaged in disaster risk reduction, specifically in the following aspects:

##### **1. Identification of disasters and their risks**

##### **2. Analysis of disaster risk**

##### **3. Treatment of disaster risk**

##### **4. Monitoring of disaster risk**

##### **5. Evaluation of disaster risk**

- The following are information checklists that CPDRRR planners must know:

##### **1. Past disasters the community experienced**

##### **2. Impact of such disasters to the lives, health, and livelihood of the community**

##### **4. Resources of the community to brace, withstand, and cope with disasters**

##### **5. Other Challenges and Lessons learned based on historical data**

#### **ELEMENTS OF CPDRRR**

- The elements of CPDRRR differ on their placement in the timeline of disasters.
- The following are elements of CPDRRR: 1.
- Prevention - It is the outright avoidance of adverse impacts of hazards and related disasters.
- It may be in the form of: a.
- Mitigation - It is the lessening or limiting of the adverse impacts of hazards and related disasters.
- Adaptation - It is the adjustment in natural or human systems in response to actual or expected climatic stimuli or their effects.
- Preparedness - It is composed of knowledge and capacities developed by governments, professional response and recovery organizations, communities and individuals.
- Response - It is the provision of emergency services and public assistance during or immediately after a disaster in

order to save lives, reduce health impacts, ensure public safety, and meet the subsistence needs of the people affected.

- Recovery - It is the restoration and improvement (where appropriate) of facilities and living conditions of disaster-affected communities, including efforts to reduce disaster risk factors.

## **THE PROCESS OF CPDRRR**

- The following are the steps in order to create a disaster risk reduction plan for a specific community:

### **1. SELECTING THE COMMUNITY**

- Communities are chosen for CBDRM activities based on a variety of characteristics and criteria, but the community's risk exposure is the most essential one. -- 1 of 2 --

### **2. RAPPORT BUILDING**

- The local social interactions and power structures, significant economic groups, and the development of strong informal contacts with the local populace should all be understood after the most susceptible communities have been identified.

### **3. PARTICIPATORY DISASTER RISK ASSESSMENT (PDRA)**

### **4. COMMUNITY-BASED DISASTER RISK MANAGEMENT PLANNING**

### **5. COMMUNITY MANAGED IMPLEMENTATION**

- The community organization at the local level should implement the plan with assistance from the local government and technical and research organizations.

- The local government should facilitate and coordinate the resource mobilization and application of the community plan.

### **6. MONITORING AND EVALUATION**

#### **Key terms**

Communities - chosen for CBDRM activities based on a variety of characteristics and criteria, but the

---

## **17. Legal Framework of Philippine DRRM**

### **Legal Framework of Philippine DRRM**

#### **INTRODUCTION**

#### **WHAT IS A LEGAL FRAMEWORK?**

#### **WHAT ARE LAWS ENACTED IN RELATION TO DRRRM?**

##### **1) HYDROMETEOROLOGICAL HAZARDS**

- It is an institution which predicts typhoons that will hit the country.
- 131 (Philippine Commission) and was renamed to the Weather Bureau.
- 10962, otherwise known as the PAGASA Modernization Act of 2015.

##### **2) EARTHQUAKE and VOLCANIC HAZARDS**

- 766 was promulgated, creating the Commission on Volcanology (COMVOL).

##### **3) DISASTER MANAGEMENT**

- 1566 created the National Disaster Coordinating Council (NDCC).
- 7160, or the Local Government Code, aimed to decentralize governments by giving more authority and power to communities.
- 9514, or the Revised Fire Code of the Philippines, assigned authority to the Bureau of Fire Protection to issue regulations and rules as regards fire disasters and hazards.
- 9729, or the Climate Change Act of 2009, creates the Climate Change Commission, which monitors and evaluates all national programs and action plans to guarantee mainstreaming of climate risk considerations across all sectors.
- 1493, or the Calamity Hazard Prevention Program Act, requires PAGASA and PHIVOLCS to establish natural calamities' hazards mitigation programs for coastal areas to perform hazard monitoring, assessments, and early warnings.
- Disaster Risk Reduction Management Plan (NDRRMP) is a policy to be implemented by the Government of the Republic of the Philippines to guarantee sustainable, climate adaptive, and disaster resilient development across sectors.
- 205, or the Department of Disaster Resilience Act, mandates the creation of the Department of Disaster Resilience, advised by the National Disaster Resilience Council, to mainstream disaster and climate risk reduction against all

## **18. RA 10121**

### **RA 10121**

#### **INTRODUCTION**

- 10121, otherwise known as Philippine Disaster Risk Reduction and Management Act of 2010.
- The following are the most important provisions of said law: 1) The creation of the National Disaster Risk Reduction and Management Council (NDRRMC), as well as its local government counterparts

#### **2) The Declaration of State of Calamity**

##### **SCOPE OF R.A. 10121**

- 1) Good governance**
- 2) Risk Assessment and Early Warning**
- 3) Knowledge Building and Awareness Raising**
- 4) Reducing Underlying Risk Factors**
- 5) Preparedness for Effective Response and Early Recovery**

#### **THE NDRRMC**

##### **Reduction and Management Council (NDRRMC)**

- 2) The National Council shall be  
Chairperson for Disaster Preparedness,  
Chairperson for Disaster Response,  
Disaster Prevention and Mitigation**

- 3) One of the most important tasks of the NDRRMC is the development of a NDRRMF, which shall provide for a comprehensive, all-hazards, multi-sectoral, inter-agency and community-based approach to disaster risk reduction and management.

#### **THE DECLARATION OF STATE OF CALAMITY**

- 10121 provides that the NDRRMC shall recommend to the President of the Philippines the declaration of a cluster of barangays, municipalities, cities, provinces, and regions under a state of calamity.
- This power of the President and local executives to declare a state of calamity includes the power to lift (remove) it.